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Qualifying Examinations Completed:
2023 (Oral): International Trade and Industrial Organization

Dissertation Title: *Essays in International Trade and Industrial Organization*

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Education:
Ph.D., Economics, Yale University, 2027 (expected)
M.Phil., Economics, Yale University, 2024
M.A., Economics, Yale University, 2024
M.A., Economics, University of Tokyo, 2021
B.A., International Relations, University of Tokyo, 2019 (minor in French Studies)

Fellowships, Honors and Awards:
Sylff Fellowship, 2024–2026
Nakajima Foundation Scholarship, 2021–2025
Research Fellow, Japan Society for the Promotion of Science (DC1), 2021

Teaching Experience:
Yale University
Fall 2025, Teaching Assistant to Prof. Costas Arkolakis, The Economics of Space
Spring 2025, Teaching Assistant to Prof. Ilse Lindenlaub, Intermediate Macroeconomics

Fall 2024, Teaching Assistant to Prof. Costas Arkolakis, The Economics of Space

University of Tokyo
Spring 2021, 2020, Teaching Assistant to Prof. Konstantin Kucheryavyy, International Trade
Fall 2020, Teaching Assistant to Prof. Yoshito Takasaki, Development Economics
Spring and Fall 2020, Lecturer, Math Camp: Optimization Theory

Research Experience:

Research Assistant to Prof. Costas Arkolakis, Yale University, 2023–2024
Research Assistant to Prof. Taiji Furusawa, University of Tokyo, 2019–2021
Research Assistant to Prof. Hideki Hashimoto, University of Tokyo, 2019–2021
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Research Assistant to Prof. Hidehiko Ichimura, University of Tokyo, 2018

Work Experience:

Economist Intern, Amazon, 2026

Publications:

"Projecting Prevalence of Frailty and Dementia and the Economic Cost of Care in Japan from 2016 to 2043: A Microsimulation Modelling Study" (2022), with Megumi Kasajima et al., *The Lancet Public Health*, 7(5), 458–468.

"The Impacts of AI, Robots, and Globalization on Labor Markets: Analysis of a Quantitative General Equilibrium Trade Model" (2022), with Taiji Furusawa and Yoichi Sugita, in *Robots and AI: A New Economic Era*, eds. Lili Yan Ing and Gene M. Grossman, Oxon and New York: Routledge.

Working Papers:

"Minimum Wage, Trade Policy, and Informality: Evidence from Colombia" (2026), *Job Market Paper*.

"Welfare Gains from Product and Process Innovations: The Case of LCD Panels, 2001–2011" (2025), with Mitsuru Igami, Jeff Qiu, and Tuyetanh L. Tran. Revise and Resubmit, *Econometrica*.

"New Technology Diffusion: Implications for Monopsony Power, Markups, and Labor Share" (2022), with Tetsuji Okazaki, Ken Onishi, and Naoki Wakamori. Revise and Resubmit, *The Review of Economic Studies*.

"Patents as Indicators of Innovation: Large Language Models Meet Ground Truth in the LCD Industry" (2026), with Mitsuru Igami, Simon Scheidegger, and Anna Smirnova.

"Markdowns and Trade Liberalization in Developing Countries: Evidence from Colombia, 1977–2020" (2024).

Seminar and Conference Presentations:

2025: Econometric Society World Congress
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2023: North American Winter Meeting of the Econometric Society (ASSA); NBER Conference "Megafirms and the Economy"; North American Summer Meeting of the Econometric Society; European Summer Meeting of the Econometric Society
2022: European Summer Meeting of the Econometric Society
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Languages: Japanese (native), English (fluent), French (intermediate), Spanish (beginner)
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Dissertation Abstract

Minimum Wage, Trade Policy, and Informality: Evidence from Colombia [Job Market Paper]

Over the past decades, many developing countries have lowered trade barriers and raised real minimum wages while informality has remained widespread. While an extensive literature examines how each policy affects informality, academic research and policy debates typically treat them in isolation. Yet the same firms and workers face both policies. Trade policy changes firms' sales and labor demand, whereas the minimum wage restricts wage adjustment in formal jobs. A firm's response to a trade shock may therefore depend on how tightly the minimum wage binds. This paper studies how a labor-market policy like the minimum wage interacts with the change in competition caused by trade liberalization and how the two jointly determine informality and welfare in developing countries.

I study these questions in Colombia, which has one national minimum wage but large regional differences in pay, so it binds more tightly in some local labor markets than in others. I combine industry-level tariff cuts from the country's 2010–2011 reform with pre-reform differences in minimum-wage exposure across local labor markets. A one-percentage-point larger tariff reduction raises informality by 0.8 percentage points where the minimum wage is more binding, but has no detectable effect where it is less binding.

To quantify the welfare effects of the minimum wage and trade liberalization and to evaluate alternative wage floors, I develop a spatial general equilibrium model of a small open economy with heterogeneous firms, employer wage-setting power, an endogenous choice between formal and informal production, and self-employment. In the model, a binding minimum wage raises wages and employment at intermediate-productivity firms by limiting wage-setting power but reduces employment or induces informal production at low-productivity firms and is non-binding for high-productivity firms. I theoretically show that trade narrows the intermediate range in which the minimum wage improves efficiency.

I estimate the model using Colombian household surveys, firm-level data, customs records, and an interregional input-output table. Simulating the 2010–2011 tariff reform, I find that aggregate informality remains essentially unchanged—a 0.08 percentage-point decline. The model also reproduces the empirical finding that tariff reductions raise informality where the minimum wage binds more tightly but not where it binds less tightly, although the differential response of informality to the tariff reform is not targeted in estimation. Whether trade liberalization raises informality therefore depends on how tightly the minimum wage binds in the regions most exposed to import competition. Holding trade policy fixed, aggregate worker welfare is maximized at a minimum wage approximately 56 percent of the current Colombian minimum wage. Reducing the minimum wage from its current level to the optimum raises aggregate worker welfare by 0.80 percent in consumption-equivalent terms. Relative to no minimum wage, the optimum benefits noncollege workers at the expense of college workers. The optimal minimum wage nevertheless remains positive because it limits employers' wage-setting power and redistributes welfare toward noncollege workers.

Welfare Gains from Product and Process Innovations: The Case of LCD Panels, 2001–2011
(with Mitsuru Igami, Jeff Qiu, and Tuyetanh L. Tran)
Revise and Resubmit, *Econometrica*

We study welfare gains from innovations in the global markets for liquid crystal display (LCD) panels. We show direct evidence on both product and process innovations by using detailed data on sales, costs, and investments. We then estimate a structural model of demand and supply to quantify their contributions. Results suggest social return on technological investment was positive, but private gains for most firms were negligible due to competition. We further investigate how competition affects firms' incentives to innovate in a series of counterfactual simulations with hypothetical mergers.

New Technology Diffusion: Implications for Monopsony Power, Markups, and Labor Share
(with Tetsuji Okazaki, Ken Onishi, and Naoki Wakamori)
Revise and Resubmit, *The Review of Economic Studies*

This paper investigates the potential implications of ignoring heterogeneity in production technology at each plant on monopsony power and markups. We collect plant-level data where we directly observe the types of production technology during new technology diffusion. We demonstrate that estimating a common production function for all plants results in upward trends in both monopsony power, measured by the differences in growth rates of MRPL and wages, and markups, obtained by production approach. However, accounting for production technology in the analysis eliminates these upward trends, highlighting the importance of controlling for heterogeneity in production technology even within a narrowly-defined industry.

Patents as Indicators of Innovation: Large Language Models Meet Ground Truth in the LCD Industry (with Mitsuru Igami, Simon Scheidegger, and Anna Smirnova)

Patents are widely used to measure innovation, but their economic content is ambiguous. We assess the usefulness of patent-based measures as indicators of the nature, incidence, and welfare impacts of innovation by leveraging a unique empirical context of the liquid crystal display (LCD) industry. We combine all US patents on LCD technology with detailed data on new products and manufacturing processes at the firm-product-fab level, as well as the structural estimates of their welfare impacts. Conventional statistics such as patent counts and citations fail to predict these ground truth events, whereas recently proposed text-based measures (e.g., novelty) exhibit some explanatory power. To extract further information from the text data, we use large language models (LLMs) to characterize and classify each patent based on its technological content and economic purpose. LLM-based measures raise the adjusted R^2 for welfare gains from near zero to 12–14 percent and for product-innovation incidence to roughly 3 percent, more than doubling the explanatory power of the best existing text-based measures. Because the labeling model has since been retired, we show that an open-weight model that anyone can download reproduces its classification quality, making the measures reproducible without proprietary API access.

Markdowns and Trade Liberalization in Developing Countries: Evidence from Colombia, 1977–2020

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This paper asks how trade liberalization changed employers' labor-market power. Using plant-level microdata for Colombian manufacturing over four decades, I estimate wage markdowns—the gap between wages and the marginal revenue product of labor—with a production-function approach. Estimated markdowns are substantial: the average plant pays wages about 15 percent below marginal revenue product. Treating Colombia's tariff reforms as quasi-experiments, I show that liberalization raised markdowns: a 1 percentage-point tariff cut raised plant-level markdowns by 0.14 percent in the 1980s reforms and by 0.2 to 0.4 percent in the 2010 reform. Markdowns are systematically larger for skilled workers, but the effect of liberalization concentrates among unskilled production workers, widening the skilled–unskilled wage gap.